

GOVERNMENT OF INDIA

NATIONAL HIGHWAYS AUTHORITY OF INDIA (NHAI)

Regional Office: Delhi-NCR · Project Implementation Unit (PIU)

Tender Reference: NIT-NHAI-NCR-2026-088	Dated: 15th January 2026
Project Name: Construction of 6-Lane Flyover & Bridge Superstructure on NH-48	Estimated Value: ₹148.50 Crores
Procurement Mode: EPC (Engineering, Procurement & Construction)	Statutory Oversight: BIS Act 2016 & CVC Guidelines

SECTION IV — TECHNICAL SPECIFICATIONS & MATERIAL COMPLIANCE

Bidders shall ensure that all supplies and materials strictly comply with the respective Indian Standards (IS Codes) and mandatory statutory orders in force.

Clause 4.1.2 — Ordinary Portland Cement (OPC) for Bridge Superstructure

All structural concrete for piers, abutments, and deck slabs shall utilize 43 Grade Ordinary Portland Cement conforming strictly to **IS 8112:1989** with minimum 28-day compressive strength of 43 MPa. The cement shall be sourced from reputed manufacturers holding valid licenses. Mandatory physical testing shall be conducted on each batch before batching plant dispensation.

[Audit Verification Note: Standard cited is IS 8112:1989. System must verify active status under Cement QCO.]

Clause 4.2.5 — High Yield Strength Deformed (HYSD) Steel Rebars

Reinforcement steel for structural columns, seismic diaphragms, and substructure elements shall be High Yield Strength Deformed (HYSD) bars Grade Fe 415 conforming to **IS 1786:1985**. Bars shall be cold twisted or thermo-mechanically treated with proof stress not less than 415 N/mm².

[Audit Verification Note: Standard cited is IS 1786:1985 Grade Fe 415. Seismic ductility requirements to be evaluated.]

Clause 6.3.1 — High Density Polyethylene (HDPE) Utility Drainage Pipes

Drainage spouts, utility conduit sleeves, and rain-water discharge piping across the deck superstructure shall utilize High Density Polyethylene (HDPE) solid wall pipes conforming to **IS 4984:1995** with PE-80 grade resin and pressure rating PN 6.

[Audit Verification Note: Standard cited is IS 4984:1995. Resin grade PE-80 vs PE-100 revision to be cross-checked.]

Clause 7.1.4 — Structural Steel Plates for Composite Girder Superstructure

Structural steel plates, flange angles, and stiffeners for composite steel girders shall conform to **IS 2062:2011 Grade E250 Quality A**. Steel shall be fully killed and manufactured by primary producers with mandatory ultrasonic inspection as per BIS test codes.

[Audit Verification Note: Standard cited is IS 2062:2011 Grade E250 Quality A.]

Clause 9.2.0 — Highway Traffic Surveillance & IP CCTV Cameras

Outdoor IP bullet cameras for automated traffic monitoring and incident detection shall provide 1080p full HD resolution, H.265 compression, minimum 30 fps, and IK10 vandal resistance with PoE interface. Test certificates from manufacturer shall be submitted.

[Audit Verification Note: Surveillance equipment procurement without explicit BIS CRS registration standard.]

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